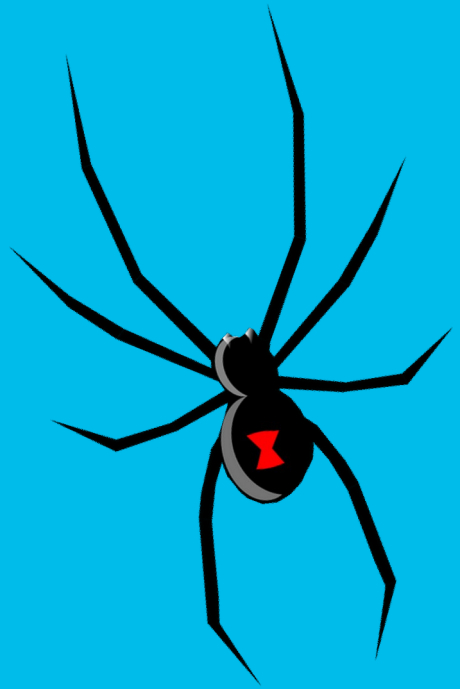


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Capstone Project

Group 5 – Black Widow

Our Group



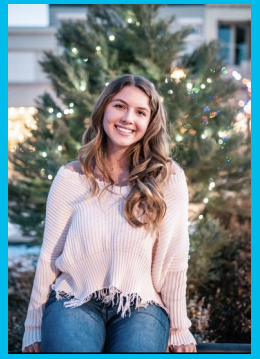
Ameer Sulyans



Edward Silva



Marcus Robles



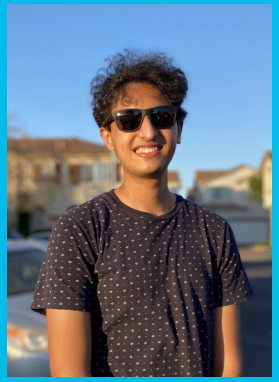
Bella Gigoux



Jeffrey Wei



Kevin Shim



Rushil Gupta



Sasha Humeniuk



Dylan Halim



Tia Kachalia

Agenda



What is the Problem?

Who we are Helping

Real World Data

Our Objective

Our Reasoning

Product Thought Process

Our Solution

Product Description

Tech Behind Product

Specifications

Our Future

Future Plans

Areas of Advancements

Questions

Ask us Anything About our Project!

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Our Question

?



How can technology be used to manage the medications adherence for chronic care patients and/or multiple medication schedules for the elderly?

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Objective

- Goal: Create a product that manages the medication schedule of geriatric patients.
- Our product will:
  - Provide notifications to consumers about their daily medications
  - Include medication security
  - Incorporate medicine accuracy
  - Enable medication dispensing per physician and pharmacology recommendations



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Geriatric Consumer's Challenges With Medication Management

- With age comes the increased need for medicine.
- Elderly grow weary and forgetful of consuming pills (e.g. Alzheimer's disease)
- Health issues including blurred vision and impaired hearing can disrupt communication between patients and care team.



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Statistics



Over 55%[1] of elderly are non-compliant with their prescription drug orders

According to the CDC, three major prescription[2] drug classes caused ~60%[3] of drug related hospitalizations

70%[4] of elderly patients who misuse prescriptions receive painkillers from more than one doctor

[1] https://www.agingcare.com/articles/medication-problems-elderly-people-have-146111.htm#:~:text=Fifty%2Dfive%20percent%20of%20the,due%20to%20adverse%20drug%20reactions.
[2] Being blood thinners, diabetes agents, and opioids
[3] https://www.agingcare.com/articles/senior-drug-overdose-148701.htm
[4] https://www.aarp.org/content/dam/aarp/ppi/2017/07/prescription-drug-abuse-among-older-adults.pdf

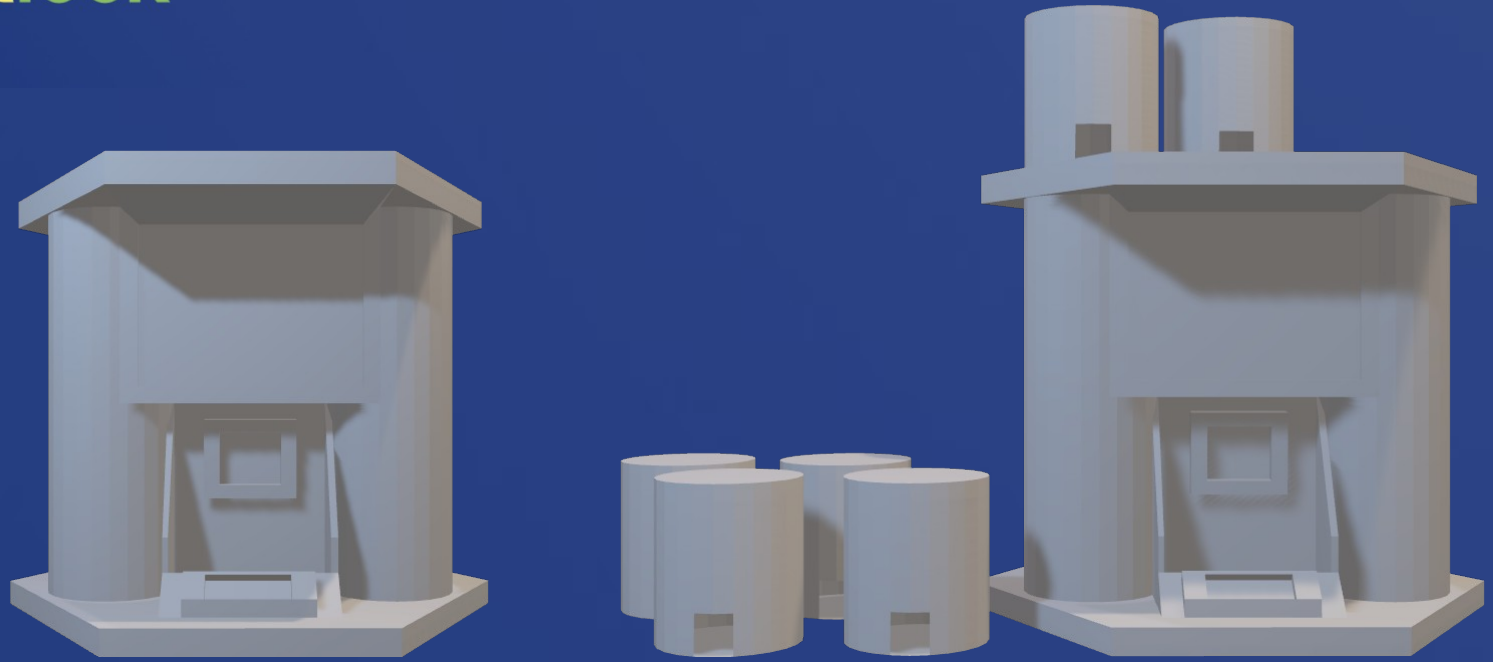
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Our Solution...

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natlock

natlock



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What is natlock?



- Reminds the patients daily to take the right amount of pills at the correct times
- Features an SOS button
- Secure because pills are locked by code chosen by care team
- Improve the quality of life for people with chronic illness

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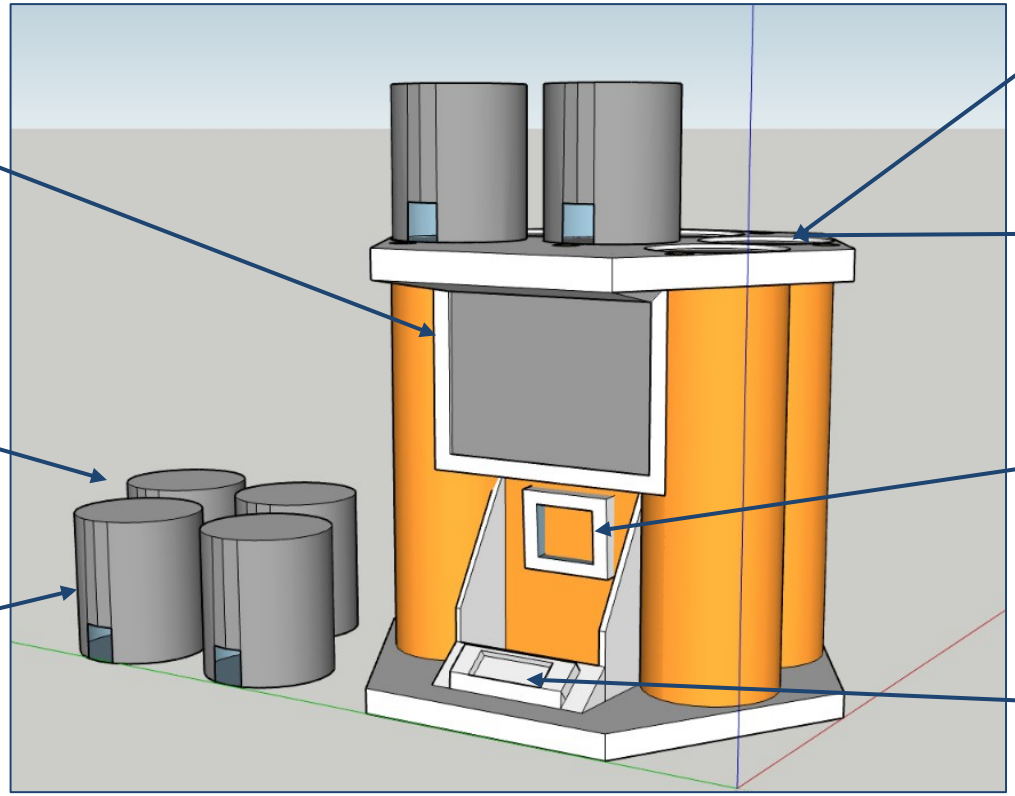
Primary Features



- Medication dispenses in accordance with prescription schedule (e.g. twice a day)
- Audio notifications via built-in speakers
- Visual Notifications via LEDs & LCD Touchscreen
- Internal environmental monitoring to preserve medication
- Emergency SOS button sends notification to care team

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Technology Behind natlock



Temperature + humidity sensor with LED

LCD Screen + Raspberry Pi

Push-to-open

6 Refillable Capsules

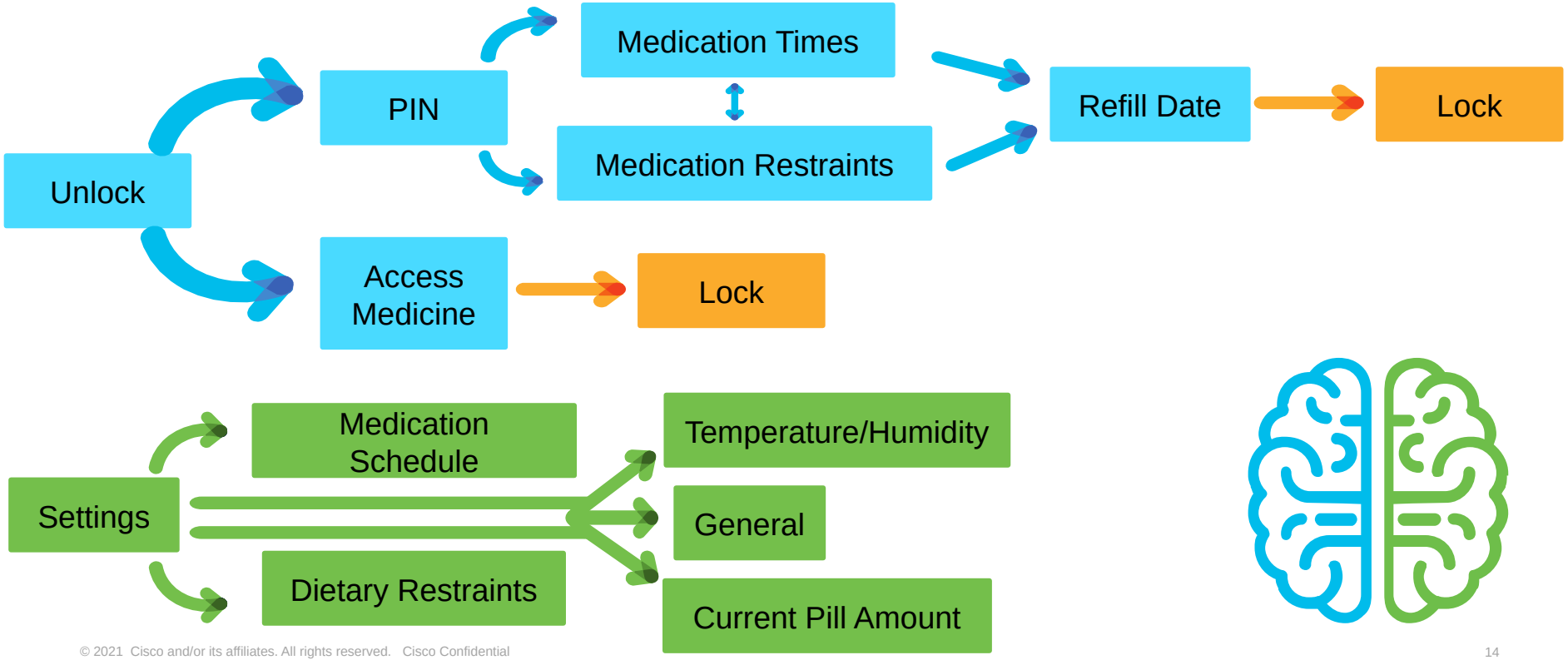
Magnet Lock Door

2 ¼” x 1” Standard Pill capsule

SOS with Protective cover On-Board GPS

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LCD Screen UI



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Energy Usage



- **natlock** uses two rechargeable D batteries.



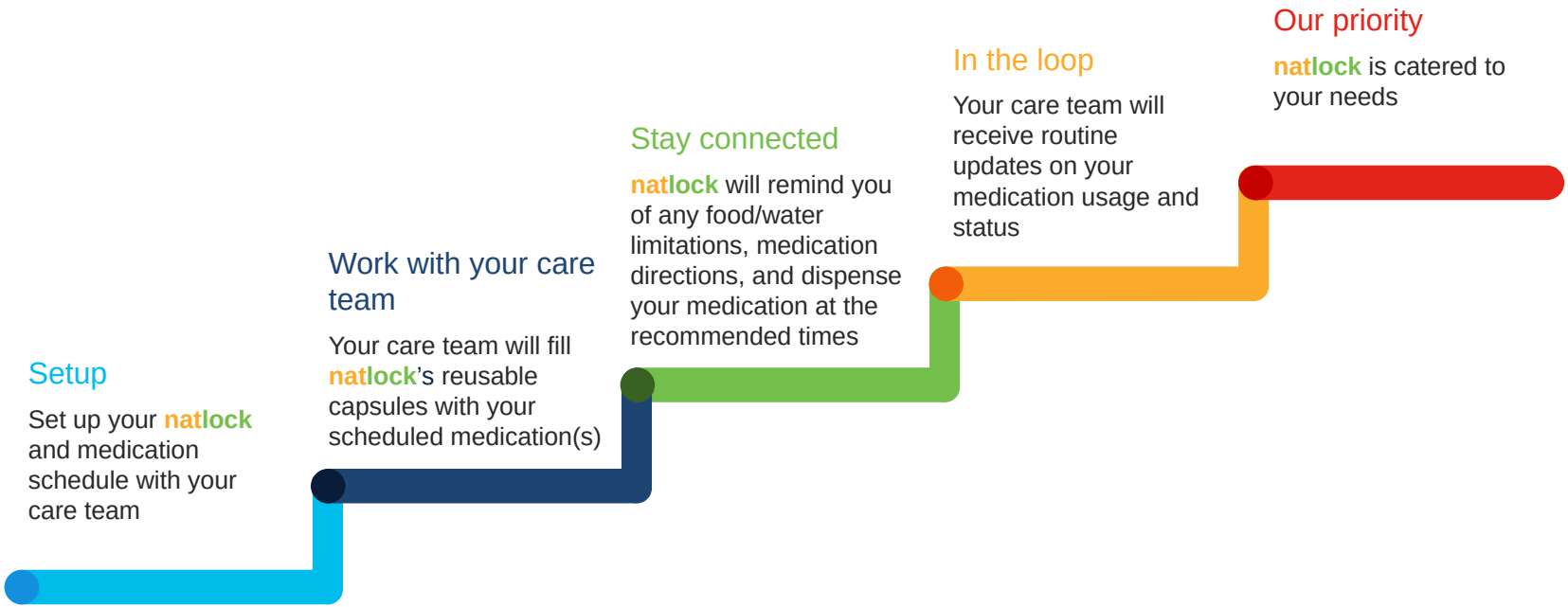
- Patients can fully charge both batteries overnight, and they can run for up to 36 hours until recharged.

- Through sound and text notifications, **natlock** will let the patient know when the batteries are running low on power ahead of time.



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How To Use natlock



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Why Us?

natlock

- GPS tracking
- Mobile, allows for travel
- Anti-Theft Technology
- SOS button, Easy to use
- Pill limit, patient can’t overdose
- HIPAA Compliance

Our competitors

- Lack of portability
- Costly subscriptions
- Technology not “elderly-friendly”
- Expensive, inaccessible
- No pill dispensation limit, overdose risk

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Future Product Improvements



- Implementing the DUO mobile app - notification system for emergency contact SOS
- Decreasing the size of the device to make it more portable
- Prolonged battery life
- Improve user interface and notification system based on feedback

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Future Platform Expansion



- Increasing accessibility for liquid and insulin doses
- Inner-capsule environment tracking
- Sustainability for more eco-friendly future
- Easier access for different socioeconomic brackets
- Multi-year warranty available
- Implement affordable subscription models to finance the product's future endeavors, such as more features

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Cost Break Down & ROI

Item	Base Price (retail) COGs/single unit	Base Price COGs @ 10% bulk discount	Customer Price/Single Unit @15% Margin
Biodegradable Filaments	$25.00	$22.50	$27.00
Raspberry Pi Zero	$5.00	$4.50	$5.40
Mini Touch Screen	$15.00	$13.50	$16.20
D Batteries	$2.50	$2.25	$3.00
Meraki Temperature Sensor	$30.00	$27.00	$32.40
Magnetic Strip	$50.00	$45.00	$54.00
GPS Sensor	$40.00	$36.00	$43.20
Speakers	$5.00	$4.50	$5.40
Sim Card	$5.00	$4.50	$5.40
3D Printer **	$20,000.00	n/a after initial investment	
Total	**$20,177.50**	**$159.75**	**$192**

Units	Investment	Profit (assuming no discounts)	Production ROI
1	$20.000.00	$32.25	0.2%
100	$20,000.00	$3,225.00	16.1%
1000	$20,000.00	$32,250.00	161.3%

COST:

- ** Industrial 3D Printer allows for mass production = more product to sell in a quicker manner/supply generation
- ** Break Even on investment is ≈621 units
- Flexibility/Cost Savings ability with different material providers

PRICE:

- Closest competitors' price range: **$150-$600**
- (Possible future) Volume Discount to customers
- Profit $32.25/unit (20% profit margin)

POTENTIAL CUSTOMERS/DEMAND GEN:

- US Population age 65+ is 40.3M* (*US Census, 2010)
- # of potential Customers/demand generation potential is high in target demographic
- Opportunity for additional end-users


Care for those
who you love





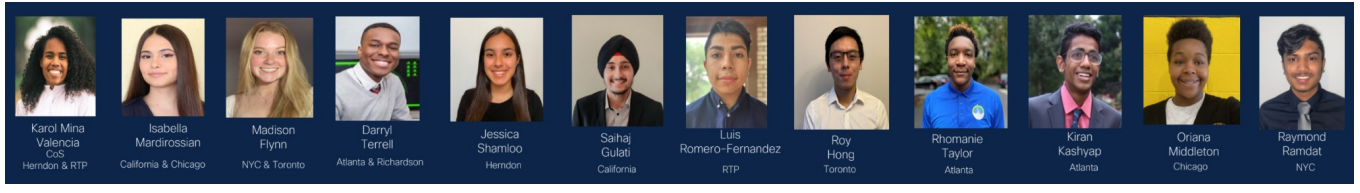
Questions?





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Thank you to all the interns and Cisco Staff for this opportunity!



Special thanks to Nicholle Pelletier, Peter Fischer, Nora Marcos, Randy Benn, Sujata Singhal, and Sweetie Browne!







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